Such An Easy Task: Detecting Sarcasm In Text

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Abstract

Sarcasm detection remains a significant challenge in Natural Language Processing due to the stark divergence between literal and intended meanings. In this study, we evaluate a novel token-level pivot-splitting strategy designed to capture intra-sentence semantic shifts against standard linear and non-linear baselines operating on sentence embeddings. Utilising a unified corpus built from three public benchmarks, models are evaluated on a binary classification task. Experimental results show that the proposed pivot-based configuration significantly underperforms, yielding a holdout F1-score of only 0.38 due to a severe collapse in recall. Conversely, a linear Logistic Regression baseline achieves the highest balance with an F1-score of 0.60, demonstrating that isolating localised context segments strips away crucial global pragmatic cues necessary for identifying sarcasm.

1 Introduction

Sarcasm serves various communicative functions, from criticism to humor and social bonding (Frenda et al., 2022; Pexman and Olineck, 2002), and manifests through diverse forms like understatement, hyperbole, rhetorical questions, and deliberate falsehoods (Leggitt and Gibbs, 2000; Riloff et al., 2013). Because its intended meaning differs from its literal interpretation, sarcasm detection is a major challenge in Natural Language Processing. Misinterpreting it distorts sentiment and opinion analysis (Frenda et al., 2022), an issue particularly critical in disaster-related contexts where monitoring public reactions on social media supports strategic decision-making (Sosea et al., 2023).

Given this complexity, we evaluate whether modelling intra-sentence semantic divergence can improve sarcasm detection. We establish a robust multi-domain benchmark by harmonising and balancing three distinct sarcasm datasets representing online debates, Twitter, and natural conversations. We then formalise a contrastive pivot-splitting architecture to explicitly model contextual shifts within a text sequence, comparing it against Logistic Regression and Multi-Layer Perceptron baselines.

Empirical evaluations on the holdout test set indicate that the proposed pivot architecture is not effective for sarcasm detection, underperforming the baseline models with an F1-score of only 0.38. Conversely, a simple linear Logistic Regression baseline operating on global embeddings yields the most balanced performance, achieving the top F1-score of 0.60. These results demonstrate that preserving global semantic information is more effective for sarcasm detection than explicitly modelling local semantic contrasts.

2 Background

Prior work in sarcasm detection has used one dataset at a time that only includes a specific style and domain of sarcasm such as social media (Abu Farha et al., 2022), online forums (Oraby et al., 2016) and TV series (Castro et al., 2019). Early sarcasm detection systems relied on handcrafted linguistic features combined with traditional machine learning models such as Logistic Regression and Support Vector Machines (Joshi et al., 2015). More recent approaches leverage distributed text representations and neural architectures, including CNNs, LSTMs, and transformer-based models, to capture semantic and contextual information more effectively (Abu Farha et al., 2022).

Despite these advances, previous studies have shown that sarcasm detection models often struggle to generalise across datasets and domains because sarcastic expressions vary substantially in style, topic, and communicative setting (Oprea et al., 2021). This motivates the exploration of how robust sarcasm detection models remain when exposed to heterogeneous sources of sarcastic language. In particular, we investigate whether explicitly modelling semantic contrasts within a sentence

can provide additional information beyond standard sentence-level representations.

3 Models

To evaluate this hypothesis, we compare three modelling strategies of increasing complexity. All models address the same binary sarcasm detection task and differ only in the way textual information is represented before classification. The input to the models is a single sentence or a paragraph written in English, and the output is a binary classification label $Y \in \{0, 1\}$, where 1 denotes a sarcastic text and 0 denotes a non-sarcastic text. To evaluate the predictive power of different textual representations, the classification task is executed over three distinct structural configurations.

3.1 Linear Baseline

The first model is a linear Logistic Regression classifier designed to provide a baseline over global features. We construct the input feature vector $\mathbf{x} \in \mathbb{R}^D$ by concatenating the global sentence embedding $\mathbf{s} \in \mathbb{R}^{384}$ extracted from the pre-trained `all-MiniLM-L6-v2` SentenceTransformer model from Hugging Face[1], and the standardised total word count N_{words}. The probability of the input text being sarcastic is modelled using a standard sigmoid function, whose exact linear formulation is detailed in Appendix A.1.

3.2 Multi-Layer Perceptron Baseline

To capture non-linear feature interactions, we deployed a Multi-Layer Perceptron (MLP) network. Let MLP : $\mathbb{R}^D \to \mathbb{R}^2$ denote the mapping function from the input feature space to the output class logits. This baseline network processes the same global feature vector $\mathbf{x} \in \mathbb{R}^D$ as the linear model through a series of hidden layers mapping to a Rectified Linear Unit (`ReLU`) (Nair and Hinton, 2010; Agarap, 2018) activation function σ and a final `softmax` (Bridle, 1990) layer. The structural recursive equations of this multi-layer mapping are reported in Appendix A.1.

3.3 Pivot-Based Multi-Layer Perceptron

While both previous models rely on a global sentence representation, they do not explicitly model potential semantic shifts occurring within a text. To address this limitation, we propose a pivot-based architecture that attempts to capture local semantic contrasts. Instead of relying solely on a global embedding, we introduce a pivot-splitting mechanism to detect abrupt contextual shifts. This process is motivated by the intuition that sarcasm is often associated with semantic incongruity within a text. For instance, a sarcastic statement frequently exhibits a sudden shift in valence or tone between its beginning and its end, where a positive remark is immediately countered by a negative or contradictory context. By systematically dividing the text sequence at strategic pivot points, we aim to isolate and measure these internal discrepancies.

Given a set of relative ratios $R \in (0, 1)$, a total word count N_{words}, and a specific ratio $r \in R$, a structural pivot index P is computed as follows:

$$P = \max(1, \min(N_{words} - 1, \lfloor N_{words} \times r \rfloor)) \quad (1)$$

The English text sequence is split at index P into a left context ($W_{1...P}$) and a right context ($W_{P+1...N_{words}}$). Both context segments are embedded independently via the pre-trained `all-MiniLM-L6-v2` model[2] to yield two dense vectors, $\mathbf{e}_{left} \in \mathbb{R}^{384}$ and $\mathbf{e}_{right} \in \mathbb{R}^{384}$.

The semantic divergence vector for each individual ratio r is isolated through vector subtraction $\mathbf{d}_r = \mathbf{e}_{right} - \mathbf{e}_{left}$. The final contrastive pivot feature vector $\mathbf{e}_{pivot} \in \mathbb{R}^{384}$ is obtained by applying an element-wise maximum pooling across all the divergence vectors computed within the set R:

$$\mathbf{e}_{pivot} = \max_{r \in R}(\mathbf{d}_r) \quad (2)$$

Classification is performed by concatenating this contrastive representation with the word count feature, yielding the final pivot feature space $\mathbf{x}_{pivot} = [\mathbf{e}_{pivot}; N_{words}]$. Let MLP : $\mathbb{R}^{385} \to \mathbb{R}^2$ denote the multi-layer perceptron mapping this updated feature vector to the final binary classification output space.

4 Experimental Setup

4.1 Data and Preprocessing

The evaluation corpus combines samples from three distinct corpora: *Sarcasm Corpus V2* (Oraby et al., 2016), *iSarcasmEval* (Abu Farha et al., 2022), and the *Conversational Sarcasm Corpus (CSC)* (Jang and Frassinelli, 2024). We select these

[1] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[2] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

sources to capture complementary linguistic domains and diverse facets of sarcasm (irony, satire, understatement, overstatement, and rhetorical questions). Appendix A.2 details their collection processes and specific limitations, while Figure 1 and Table 1 provide class distributions and qualitative text examples.

To mitigate single-domain bias, we apply a dataset unification framework that resamples each source to extract an equal proportion of instances per domain. For the *CSC* dataset, which features a 1-to-5 sarcasm intensity scale across multiple annotations, continuous scores are averaged. Following Jang and Frassinelli (2024), these mean scores are binarised using a threshold $T = 4$:

$$\text{Label} = \begin{cases} 1 & \text{if mean score} \geq 4 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

All textual instances undergo standard normalisation: conversion to lowercase, removal of punctuation and special characters, and stripping of redundant whitespace. For the *CSC* corpus, conversational context is discarded to retain exclusively the target response text.

Global semantic features are extracted via the pre-trained `all-MiniLM-L6-v2` SentenceTransformer model, yielding a dense embedding vector $\mathbf{s} \in \mathbb{R}^{384}$. Non-predictive metadata strings are completely discarded. To prevent data leakage, the standardisation of the word count feature N_{words} via a `StandardScaler` (Pedregosa et al., 2011) is integrated directly into the cross-validation pipeline, ensuring the scaler is fit strictly within each training fold.

4.2 Data Splitting and Cross-Validation

The unified dataset is partitioned into training and testing sets using a stratified $80/20$ split to preserve both class and source distributions across subsets. For hyperparameter optimisation within the training set, a 7-fold cross-validation scheme is implemented. The cross-validation splits are multi-stratified based simultaneously on the target sarcasm labels and the original dataset sources to account for both class and domain imbalances.

4.3 Training Procedure and Hyperparameters

Three distinct modelling configurations are evaluated using specific optimisation strategies:

- **Logistic Regression Baseline:** The model is trained using the `saga` solver (Defazio et al., 2014) to manage the embedding space efficiently. Class weights are adjusted dynamically using a `balanced` configuration (Pedregosa et al., 2011). A grid search is performed to optimise the regularisation strength ($C \in \{0.01, 0.1, 1, 5, 10, 50\}$) and penalty types (`l1`, `l2`, or none) based on the validation F1-score.

- **Multi-Layer Perceptron (MLP) Baseline:** The network is trained on global features. Optimisation is carried out over a grid of hidden layer topologies, specifically $(128, 64)$, $(128, 64, 32)$, $(256, 128)$, and $(256, 128, 64)$, utilizing the `ReLU` activation function. Regularisation is controlled via the L2 penalty parameter ($\alpha \in \{0.0001, 0.001, 0.01\}$). Training is limited to a maximum of 500 epochs with early stopping enabled.

- **Pivot-Based MLP:** This model shares the exact same network configurations and hyperparameter grid as the MLP baseline, but operates on the intra-sentence contrastive features $\mathbf{e}_{pivot}$. An additional layer of optimisation wraps the model selection over four relative ratio configurations: $R_1 = \{0.5\}$, $R_2 = \{0.3, 0.7\}$, $R_3 = \{0.3, 0.5, 0.7\}$, and $R_4 = \{0.1, 0.5, 0.9\}$.

4.4 Metrics and Significance Testing

Model performance is evaluated using standard classification metrics, which are described intuitively below and defined formally in Appendix A.3.

Accuracy measures the global proportion of correctly classified instances across both classes. Precision quantifies the model's exactness by calculating the ratio of correctly predicted sarcastic instances out of all instances predicted as sarcastic by the system. Recall evaluates the model's completeness by computing the proportion of actual sarcastic instances that are successfully identified. Finally, the F1-score provides the harmonic mean of precision and recall, serving as the primary optimisation metric to balance the trade-off between exactness and completeness under potential class imbalances.

To validate whether performance differences between models are statistically meaningful, a paired Monte Carlo permutation test is executed with $R = 10,000$ random shuffles. Pairwise comparisons are conducted across all model configurations

on the holdout test set. To control the False Discovery Rate (FDR) stemming from multiple testing, p-values are adjusted using the Benjamini and Hochberg (1995) correction procedure with a significance threshold set at $\alpha = 0.05$.

5 Results and Discussion

The primary and most unexpected finding is that our proposed Pivot-Based MLP significantly underperforms compared to both baselines. As shown in Table 2 and Figure 2, pairwise Monte Carlo permutation tests confirm that its lower F1-score (0.38) is highly statistically significant against the Linear Baseline ($\Delta = 0.22$, $p < 0.001$) and the MLP Baseline ($\Delta = 0.12$, $p < 0.001$), a result further illustrated by the significance matrix (Figure 3). These results do not support our initial hypothesis that isolating intra-sentence semantic divergence would implicitly capture sarcasm.

An ablation and error analysis reveals that this failure stems from a severe drop in recall (0.27), meaning over 72% of sarcastic statements are incorrectly classified. As visible in Table 2, the Pivot-Based MLP produces an excess of False Negatives. By segmenting the text and discarding global embeddings, the model becomes structurally blind to pragmatic cues, adopting an overly conservative strategy that labels almost all ambiguous inputs as non-sarcastic.

Conversely, the Logistic Regression baseline achieves the best balance, maximising the F1-score at 0.60. Although the MLP Baseline reaches a higher Accuracy (0.72) and Precision (0.60) than the linear model (0.70 and 0.53), its recall remains restricted (0.43). Interestingly, the F1-score difference between the two baselines ($\Delta = 0.03$) is not statistically significant ($p = 0.05$, see Table 3). This implies that applying non-linear multi-layer topologies to global sentence embeddings provides no statistically verifiable advantage over a simple linear combination for balanced sarcasm detection.

6 Conclusion

In this study, we investigated the effectiveness of a token-level pivot-splitting mechanism designed to isolate intra-sentence semantic divergence for binary sarcasm detection in English texts. While our initial hypothesis anticipated that localised contextual shifts would serve as a strong proxy for sarcastic incongruity, empirical results on a multi-stratified holdout test set invalidated this assumption. The proposed Pivot-Based MLP model underperformed significantly, failing to identify over 72% of sarcastic statements and achieving a restricted F1-score of 0.3801. In contrast, the linear Logistic Regression baseline operating on sentence embeddings from `all-MiniLM-L6-v2` provided the most robust and balanced performance, securing the highest F1-score of 0.60.

These findings suggest that text-based sarcasm detection relies heavily on global, multi-layered pragmatic cues that are disrupted when a text sequence is segmented into isolated left and right context blocks.

7 Limitations

Several limitations inherent to our data and task formulation must be addressed to explain our models' modest performance and guide future improvements:

- **Annotation Noise:** Inconsistent labels within the source datasets introduce significant noise. For instance, in the *CSC* corpus, formulaic expressions like "Cool story bro" are classified as sarcastic, whereas "Cool." is not, which confuses the models during training.
- **Omission of Context:** Sarcasm relies heavily on situational cues. By discarding preceding conversational turns to maintain feature uniformity, we stripped away vital pragmatic information, directly causing the severe collapse in model recall.
- **Typological Homogenization:** We collapsed highly diverse rhetorical forms (irony, hyperbole, understatement) into a single binary framework. This homogenisation obscures stylistic nuances, making it difficult for models to generalise across conflicting types of sarcasm.

To improve these scores, future work should integrate contextual embeddings from preceding dialogue turns and transition toward multi-task learning frameworks to explicitly distinguish between different sarcasm types.

8 Code Availability

The code implemented in this study is publicly available at: `https://github.com/Lenarbrs/2026_Sarcasm_Detection`.

9 Statement

- **Éloïse Delerue** contributed to the literature review, developed the initial framework of the codebase utilised for the poster presentation and contributed to the drafting and editing of the manuscript.
- **Lucile Lapray** contributed to the literature review, designed the visual materials for the poster presentation, and implemented the majority of the post-presentation code.
- **Léna Rebours** contributed to the literature review, executed the dataset collection and unification framework, developed the statistical significance testing pipeline, and contributed to the drafting and editing of the manuscript.

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A Appendix

A.1 Mathematical Formulations of Baseline Models

Logistic Regression Framework Let $\mathbf{x} \in \mathbb{R}^D$ represent the concatenated input feature vector, $\mathbf{w} \in \mathbb{R}^D$ denote the model weight vector, and $b \in \mathbb{R}$ represent the scalar bias term. The formal probability distribution of the binary classification layer is formulated as:

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\mathbf{w}^T\mathbf{x} + b))} \quad (4)$$

MLP Hidden Layer Recursion For the baseline Multi-Layer Perceptron network, the mapping functions pass through successive hidden dimensions. Let $\mathbf{W}^{(l)}$ denote the weight matrix for a given layer l, and $\mathbf{b}^{(l)}$ represent its corresponding bias vector. Given the initial feature distribution $\mathbf{h}^{(0)} = \mathbf{x}$, the network hidden representations are updated recursively as follows:

$$\mathbf{h}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}) \quad (5)$$

where σ corresponds to the element-wise activation mechanism. The final layer logits are projected via an explicit softmax operations layer to determine the class probabilities over the 2-dimensional output space.

A.2 Detailed Dataset Descriptions and Properties

Sarcasm Corpus V2 The *Sarcasm Corpus V2* (Oraby et al., 2016) is extracted from the larger Internet Argument Corpus and contains 9,386 posts sampled from online debate forums. Approximately half of these instances are labeled as sarcastic, as illustrated in the class distribution plot (see Figure 1). Each text instance was evaluated by multiple annotators who had access to both the target response and its conversational context, ensuring high inter-rater agreement.

However, this corpus presents notable structural limitations. The officially released version does not include the original preceding conversational contexts. Furthermore, because the source text stems exclusively from highly argumentative debate forums, it introduces specific linguistic and stylistic biases that may not generalize well to spontaneous, everyday communication.

iSarcasmEval The *iSarcasmEval* dataset (Abu Farha et al., 2022) consists of 3,801 English Twitter posts, which include 1,067 sarcastic examples. A unique and valuable characteristic of this dataset is that the sarcasm labels were provided directly by the authors of the tweets themselves. This guarantees that the annotations reflect the true, intended sarcastic meaning of the speaker rather than a third-party interpretation.

Netheverless, *iSarcasmEval* lacks any contextual or situational metadata surrounding the tweets. It also exhibits a significant class imbalance, with non-sarcastic instances heavily outnumbering sarcastic ones, as shown in Figure 1.

Conversational Sarcasm Corpus (CSC) The *Conversational Sarcasm Corpus (CSC)* (Jang and Frassinelli, 2024) is gathered from naturally occurring, offline face-to-face conversations, capturing sarcasm in a realistic, multi-turn dialogue setting. It provides rich contextual information and includes both author-based and third-party annotations.

We note a major structural constraint regarding this dataset: sarcastic examples constitute only a very small proportion of the overall natural corpus, creating a severe class imbalance (see Figure 1). Furthermore, although the original corpus is highly valuable due to its rich contextual metadata, we discard this conversational context to maintain feature uniformity with the other two non-contextual baselines, evaluating all models strictly on the target response text.

A.3 Mathematical Formulations of Evaluation Metrics

Let TP, FP, TN, and FN denote the number of True Positives, False Positives, True Negatives, and False Negatives, respectively. The classification metrics used to evaluate the models are mathematically formulated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

A.4 Supplementary Tables and Figures

Dataset Source	Label	Text Sample
Conversational Sarcasm Corpus (CSC)	Non-sarcastic (0)	"I'd love to go to Japan, where do you recommend?"
	Sarcastic (1)	"If you don't play, think of all those kids with cancer you've indirectly killed"
iSarcasmEval	Non-sarcastic (0)	"Pub crawl pub crawl pub crawl pub crawl pub crawl. There are pubs and I am crawling too them"
	Sarcastic (1)	"If you don't like Lemonade you're an extremist"
Sarcasm Corpus V2	Non-sarcastic (0)	"Because there are still a lot of people in this country that support oppression and injustice toward gay people."
	Sarcastic (1)	"Aaahhh, so just not accomplishing our goals and going home is not defeat, there has to be paperwork for it to be a defeat. Gotcha."

Table 1: Qualitative examples of sarcastic and non-sarcastic text samples across the three unified dataset sources.

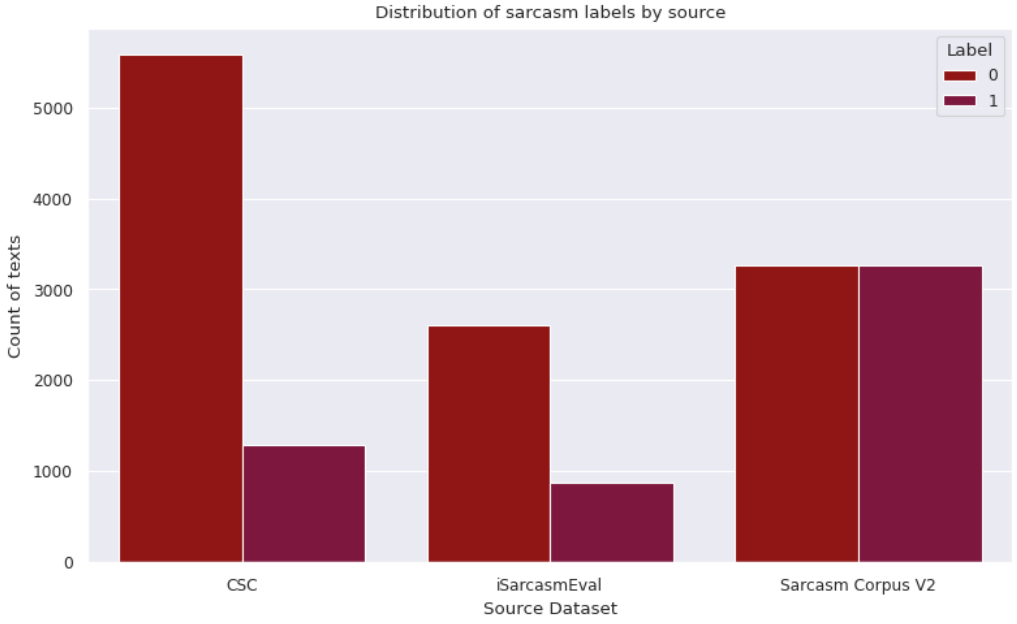


Figure 1: Distribution of sarcastic and non-sarcastic instances across the three source datasets before unification.

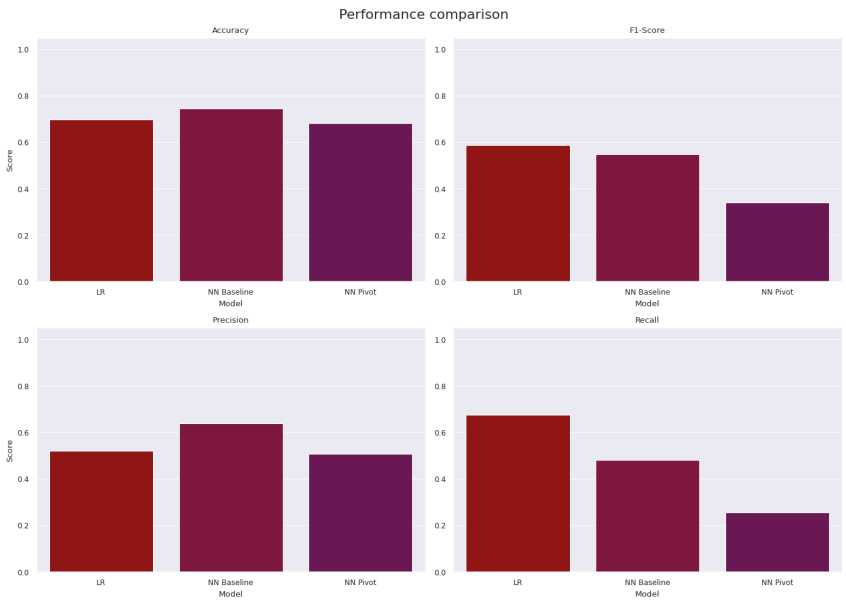


Figure 2: Graphical performance comparison across accuracy, precision, recall, and F1-score.

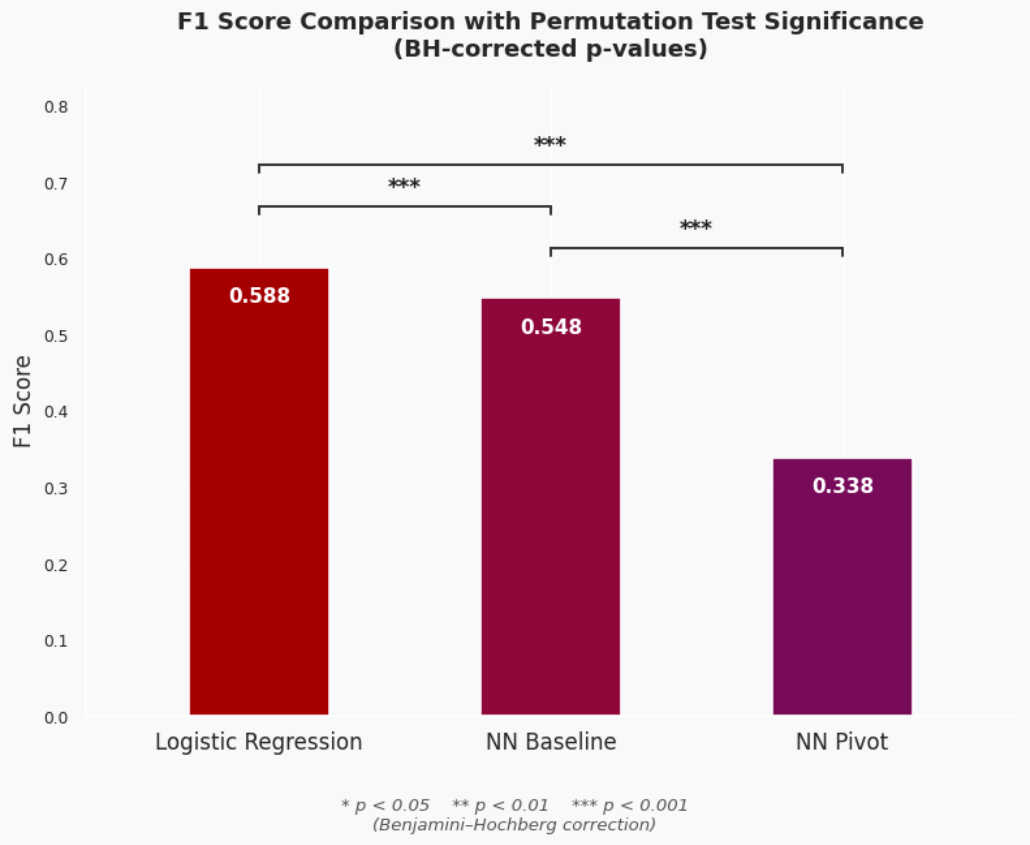


Figure 3: Pairwise comparison of F1 scores using permutation testing with Benjamini–Hochberg correction. All pairwise differences are significant at $p < 0.001$.

Model	Acc.	Prec.	Rec.	F1
Logistic Regression	0.6970	0.5312	0.6799	0.5964
NN Baseline	0.7178	0.5980	0.4305	0.5006
NN Pivot	0.7072	0.6253	0.2730	0.3801

Table 2: Performance metrics on the held-out test set.

Comparison	**Δ F1-score**	***p*-value (raw)**	***p*-value (BH)**	**Significance**
Logistic Regression vs. MLP Baseline	0.0267	0.0526	0.0526	ns
Logistic Regression vs. Pivot-Based MLP	0.2198	0.0001	0.0003	***
MLP Baseline vs. Pivot-Based MLP	0.1206	0.0001	0.0001	***

Note: *** indicates highly statistical significance at the $p < 0.001$ level after adjustment; *ns* indicates no statistically significant difference ($\alpha = 0.05$).

Table 3: Pairwise model comparisons using a paired Monte Carlo permutation test ($R = 10,000$) with Benjamini-Hochberg (BH) correction on the holdout test set.